Academic Statement of Purpose

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Over the course of my training at IISER Mohali, I have explored diverse areas of mathematics, including differential geometry, dynamical systems, symplectic geometry, complex geometry, representation theory, Lie groups, ergodic theory and mathematical physics. I have had the opportunity to credit courses on knot theory, Riemannian geometry, elliptic curves, Fourier analysis in my major years at IISER Mohali. Still, I believe the course I benefited from the most was the one on curves and surfaces in third semester, which led to my interest in the theory of manifolds and algebraic topology quite early.

Engaging in above mentioned subjects led to me inculcate a diverse set of theoretical and problem solving skills, which firstly lets me engage in different conversations quickly. And in turn conversing with my peers lets me learn and relearn new and old ideas and skills. I have had fun giving talks for the mathematics club, courses, graduate seminar etc.

1 My path to the ergodic theory at infinity

As I had a little background in algebraic topology and had credited courses on Riemannian geometry, I could recover the *pre-moduli* classification of Riemann surfaces assuming the theorem of uniformization of simply connected Riemann surfaces in terms of quotients of these simply connected Riemann surfaces by discrete subgroups of their automorphism groups. This led me to discover the Teichmuller and moduli spaces of smooth surfaces. This geometric and algebraic aspects of the same object I find recurring in my thesis work.

1.1 Ergodic theorems on actions of locally compact amenable groups

For a summer reading under Dr. Jotsaroop Kaur, motivated from the transference principle in harmonic analysis, I started looking at geometric and volume growth aspects of groups along with what ergodic properties do actions of such groups have. I presented amenability of compact and Abelian groups and the proof of ergodic theorems on actions of locally compact amenable groups with some volume growth conditions.

These notions led me to discover the fascinating world of geometric group theory from Clara Loh's book and the theory of locally symmetric spaces and arithmetic subgroups of semi-simple Lie groups from Dave Witte Morris's book and more broadly the text on discrete subgroups of Lie groups by Raghunathan. I still did not know how ergodic theory may appear in this geometric and group theoretic world that I had discovered.

A workshop on the rigidity of discrete groups

Later in the summer, I attended a workshop on the *rigidity of discrete groups* held in IISER Mohali, orgnanized by Dr. Parab Sardar and Dr. Krishnendu Gangopadhyay. I was fascinated by the theorems on rigidity of hyperbolic 3-manifolds and some complex manifolds, having seen the theory of *deformations* in dimension 2.

A seminar on the boundary of symmetric spaces

I became a part of a seminar on the boundary of (globally) symmetric spaces, organized by my thesis advisor Dr. Arghya Mondal, which has been a new experience for me! To work around some pre-requisites, we started with the CAT(0) symmetric space structure on the set of all positive definite real matrices $P(n, \mathbb{R})$. I presented the embedding of symmetric spaces of non-compact type inside $P(n, \mathbb{R})$ as complete and totally geodesic submanifolds and thus reducing our work to such submanifolds. Then we constructed the visual and Tits boundary of such symmetric spaces.

Patterson-Sullivan densities on the limit set at infinity

Eventually, I decided to do my masters thesis on the rigidity of discrete subgroups of $\mathrm{Isom}(\mathbb{R}\boldsymbol{H}^n) \cong O^+(1,n)$. I started looking at Patterson-Sullivan densities supported on the limit set of such discrete groups. We follow Sullivan for a proof of a certain rigidity theorem for discrete groups of divergent type: which generalizes Mostow's rigidity theorem.

I quickly found the results on limit sets, ergodic actions, quasiconformal maps needed for the main theorem. This brought together a lot of tools that I had been learning about for the past four years. My skills in measure theory, complex analysis and Fourier analysis came handy while dealing with quasiconformal maps. I picked up some theory of hyperbolic 2-manifolds and 3-manifolds so that I could have examples of the discrete groups I was working with: the courses on Fuchsian groups and Knot theory helped immensely in this regard.

Slowly, I made progress on understanding Sullivan's proof of the main theorem. I look at multiple papers talking about the same proof, using it in varying levels of generality. The comments from such papers helped me understand why each step works the way it does. I am trying to fix missing steps on the original proof. I have found myself thinking about it while talking a walk and coming up with a possible solution, then testing it, but then realising it doesn't work. So I try to find possible solutions for a special case and then trying to generalize it to a broader class of objects. Eventually, bit by bit I made progress on the main result, showing divergent discrete group of $\mathrm{Isom}(\mathbb{R}\boldsymbol{H}^n)$ are measurably rigid!

It still shocks me to think how so many different ideas, ranging from hyperbolic geometry to analysis of weakly differentiable functions on the sphere, come together for the such proofs. I also looked at multiple proofs of Mostow rigidity and was shocked to find even more variety of ideas in the different proofs!

2 Experiments with talks on symplectic and complex manifolds

When I was in third year, I was interested in Hamiltonian flows and topological properties of such flows. I wanted to give a talk proving that orbits in irrational translations on the 2-torus are dense in the torus. This was an exercise in the texbook by Perko. Boiling down the argument to density of orbits in irrational circle rotations was easy. However, after that I tried to prove the orbit of 1 in the circle under a irrational rotation is a subgroup which is not finite, hence it must be dense. This turned out not so well for the audience who had little experience with compactness. I recently discovered very simple proofs of the proposition about irrational circle rotations just using the fact that it is an isometry of the circle!

Slowly reading Ana Cannas da Silva's text on symplectic geometry over the course of a year, I gave a talk on the introduction to symplectic manifolds for a graduate seminar in IISER Mohali. I wanted to motivate the very definition of a symplectic manifold as the "minimum" structure on a smooth manifold needed to define Hamiltonian vector fields of smooth functions. Therefore, I first started with Hamiltonian flows on the plane: where Hamiltonian vector fields are rotation of the gradient vector field

$$J\mathrm{grad}(H)$$

and sketched the proof of one-to-one correspondence of smooth functions and area preserving flows on the plane. This was met with appreciation from the audience!

Then I moved onto the 2-sphere: because I find the symplectic structure of the sphere in terms of the Riemannian metric and the complex structure (using cross product) a good motivation for declaring: a symplectic structure is all you need to replace the complex structure and the Riemannian metric and still have Hamiltonian flows (of $H \in \mathcal{C}^\infty(M)$) which are "volume preserving" and also "preserves" the function H.

Recently, I looked at the first few sections of the text by Helmut Hofer and Eduard Zehnder on *Symplectic Invariants and Hamiltonian Dynamics*. It fascinated me how a hypersurface $S \subset \mathbb{R}^{2n}$ carries a prescribed collection of orbits without any choice of a Hamiltonian vector field!

Another talk I gave in fourth year was for the course on Arithmetic of elliptic curves. I had to prepare something related to the topics discussed in class, and we had discussed the Weirstrass function parametrizing the elliptic curve. I was interested in looking at how integrating a holomorphic 1-form gives inverse of this parameterization by the complex 2-torus. The problem I faced was: where should I begin? Should I assume the results proved in class? If yes, then the proof became an one liner (present in the textbook Arithmetic of elliptic curves). Therefore, I choose to redo the entire theory, starting from the proof that elliptic curves are homeomorphic to 2-tori, so that I can get their fundamental group to have two generators. I did this by explicitly constructing charts for the elliptic curve $y^2 = x(x-1)(x-\lambda)$ by using the function

$$x \mapsto (x, \pm\sqrt{x(x-1)(x-\lambda)})$$

This helped in another aspect: showcasing how the integral of the holomorphic 1-form

$$\frac{\mathrm{d}x}{y} = \frac{\mathrm{d}x}{\sqrt{x(x-1)(x-\lambda)}}$$

can be made very explicit! After that it was simple exercise to show the subset Λ generated by integra-l of this 1-form on the fundamental group is a rank 2 lattice, and that the integral gives a well-defined holomorphic bijection onto $\mathbb{C}/\Lambda$.

This was the genus = 1 case of the theorems by Abel and Jacobi. In the summer, I wanted to prove the general genus case next. Me and my friends started a seminar series, with introductory talks by Prof. Kapil Hari Paranjape, on compact Riemann surfaces. We presented sketch of proofs of uniformization, Riemann-Roch and Abel-Jacobi theorems.

3 My background and interests in differential geometry

After my encounter with differential forms in algebraic topology and gaining a background on the theory of smooth manifolds, the setup of mechanics using symplectic manifolds started to make sense to me. I started reading parts of Ana Cannas da Silva's text on symplectic geometry. The first de Rham cohomology popped up as a topological obstruction to Noether's theorem in the Hamiltonian side.

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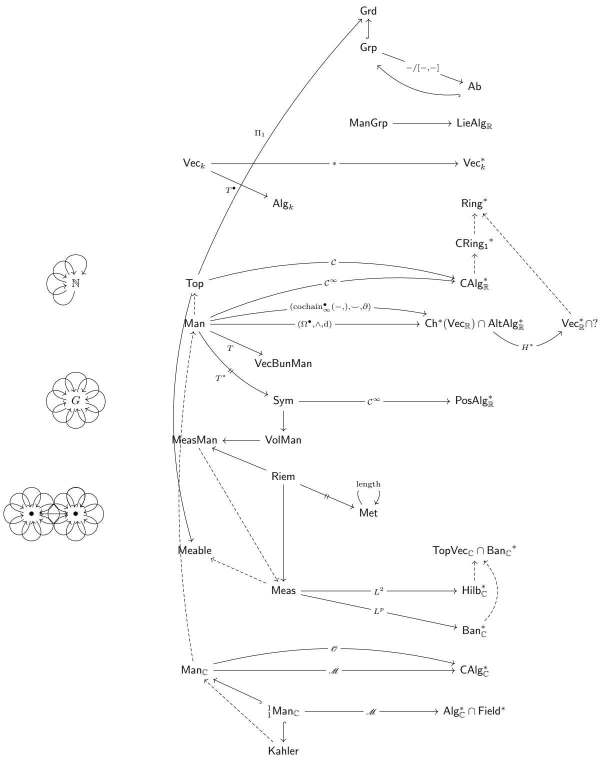
My interests in mathematical physics such as quantum field theory, led to the field of gauge theory which is where I discovered the tangled theories of integrable systems, gauge theories, topology and physical mathematics. I started reading Hitchin's book on the complex algebraic geometry of Lax pair equations, and I heard about the conjectures and works by Hitchin and Donaldson on gauge theories that have connections to integrable theories as well.

I have been fascinated with complex manifolds, such as the theory of vector bundles on complex manifolds, hodge theory, etc., however unfortunately I have not got a chance to study them. Mirror symmetry and Calabi-Yau manifolds remain another distant goal for me.

4 View on mathematics

My view on mathematics mainly stems from my background in geometry, topology and dynamics.

I view the various structures studied in mathematics as a quiver: categories of structures and functors between them (thus, living in the category of cateogries Cat). A group action is thus just a functor from the group (viewed as a category) to the category of sets Set. We may consider such a functor going into Top, and then compose with various functor coming out of Top such as π_1 and get *invariants*. Many such functors arise from the correspondence between geometry of spaces and algebra of functions on them.



On the other side I see the various explicit objects of study such as $\mathbb{R}^n, \mathbb{C}^n, \mathbb{R}H^n, \mathbb{C}P^n$, and functions on them, various other explicit Riemannian manifolds, groups, flows, group actions, etc.

Working in mathematics for me is a tug of war of information between these two worlds. We start with some explicit object and extract the correct structure out of it, which in turn gives us more explicit objects when we try to classify them. This back and forth, is what I picture when I write proofs classifying objects with some structure. For instance, it still fascinates me how proving existence of a counterexample of a statement implies we will never be able to write a (correct) proof of the statement.

My thesis work has led me to a place in between of these two worlds. Cartan had already classified (globally) symmetric spaces of non-compact type, and its geometry is also well-understood. Their isometry groups, as in semi-simple Lie groups of non-compact factors are also well-studied. Thus, quotients of such spaces, locally symmetric spaces, are more or less explicit - but they have large moduli spaces in the case of dimension 2, and fascinating properties like rigidity and superrigidity in other cases.

5 Future prospects

I wish to start my graduate studies in 2026. Given my background and skills, I feel I am equipped to pick up and understand a given topic in differential geometry and related areas.
I wish to join a graduate school in mathematics to continue my study of phenomena in these fields and

- foster broader connections with the mathematical community,
- teach and discuss mathematics with beginners,
- study the topics that left me fascinated

and eventually contribute to mathematics meaningfully.